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DATA COLLECTION

1. Objective:

To develop and validate a Python-based data collection pipeline that gathers real-time weather and air quality data for multiple locations in Bengaluru, and concurrently scrapes structured data from an online bookstore for potential integration into environmental, behavioral, or commercial analysis.

2. Descriptions:

This experiment combines three types of data sources into a single automated workflow:

Weather Data

Collects hourly temperature and relative humidity for five Bengaluru locations over the past 21 days via the Open-Meteo API.

Air Quality Data

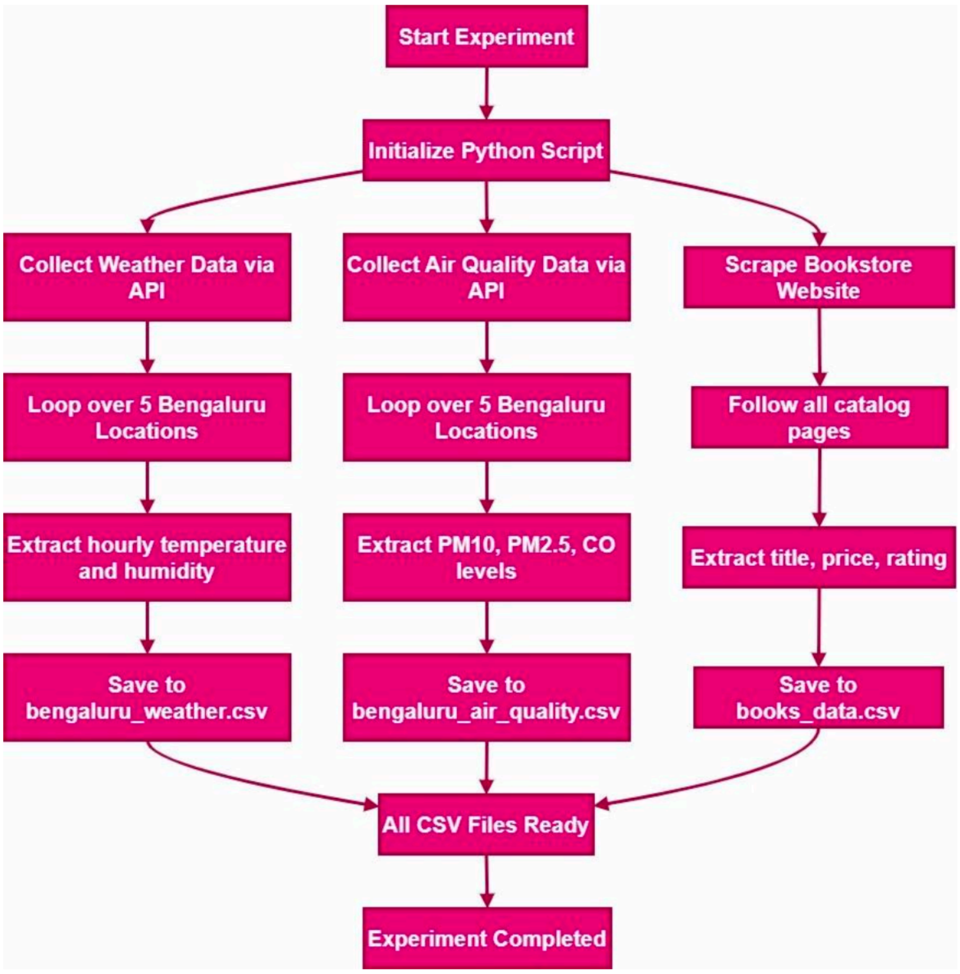
Fetches hourly pollutant concentrations (PM10, PM2.5, CO) for the same five locations using the Open-Meteo Air Quality API, covering a 30-day period.

Bookstore Data

Uses web scraping techniques to extract book title, price, and rating from a practice online bookstore across multiple paginated catalog pages.

The gathered data is stored in structured CSV files for downstream analytics or modeling purposes.

3 . Model or Flowchart or Architecture



4. Algorithm

1. START
2. Initialize required Python libraries:

   :- requests, pandas, time, BeautifulSoup, csv, etc.
3. Define a list of 5 Bengaluru location coordinates (latitude and longitude).

4. For each location in the list:

    4.1 WEATHER DATA COLLECTION

    a. Formulate the Open-Meteo weather API URL with current location.

    b. Send HTTP GET request to the API.

    c. Parse JSON response to extract:

    - Hourly temperature

    - Hourly humidity

    d. Append the data to a weather DataFrame.

    4.2 AIR QUALITY DATA COLLECTION

    a. Formulate the Open-Meteo AQ API URL with current location.

    b. Send HTTP GET request to the AQ API.

    c. Parse JSON response to extract:

    - PM10, PM2.5, and CO levels

    d. Append the data to an air quality DataFrame.

5. After looping through all locations:

    5.1 Save the weather DataFrame to `bengaluru_weather.csv`.

    5.2 Save the air quality DataFrame to `bengaluru_air_quality.csv`.

6. BOOKSTORE WEB SCRAPING

    a. Define the base URL of the bookstore.

    b. Initialize an empty list to store book data.

    c. While next page exists:

    i. Send HTTP GET request to the page.

    ii. Parse HTML using BeautifulSoup.

    iii. For each book on the page:

    - Extract title, price, and rating.

    - Append the data to the list.

    iv. Identify link to the next page.

7. Save the scraped book data to `books_data.csv`.

8. END

5. GitHub Link:

https://github.com/HA25HO/Data-Science-